

[0045] By storing all of these identified dimensions, the data graph can be quickly and efficiently visualized and manipulated. For example, rather than responding to a user query to filter nodes based on a particular dimension by searching one or more data sources to locate the relevant data and mapping the relevant data to the data graph as part of responding to the query, the relationship information included in the stored data graph allows a response to be provided to such a user query quickly and with more efficient use of computing resources and communication bandwidth.

[0046] As noted above, in some embodiments, the electronic processor 300 may repeat the expansion process for a data graph stored in the data graph database 125 periodically or as new data becomes available. For example, the electronic processor 300 may be configured to repeat the above described expansion process when an organization involved in the production or distribution of the entity provides new or updated information, a new data source becomes available or accessible by the electronic processor 300, or the like.

[0047] As described above, to allow for different visualizations or renderings of the data graph, the electronic processor 300 may be configured to create one or more separate visualization layers for a created data graph. For example, in some embodiments, the electronic processor 300 (when executing the visualization layer creation software 325) creates a visualization layer for one or more types of visualizations or renderings available for a data graph (which may be defined for all data graphs or based on the data or structure of the data graph). Each visualization layer may provide a standard way of defining data included in the data graph for a particular type of visualization or rendering. For example, one of the available visualizations may include a two-dimensional or three-dimensional map and the visualization layer for this type of visualization may map data included in the data object to locations (positions) on the map. Similarly, for a visualization that provides a timeline view, the visualization layer may map or tag data included in the data graph to particular locations or aspects of the timeline. Thus, the visualization layers define features for plotting or mapping data included in the data graph to a particular type of visualization or rendering. Accordingly, the visualization layers can represent cached views including metadata defining how data included in a particular data graph can be visualized and explored. As noted above, creating the visualization layer separate and distinct from the data graph allows the data graph to be generated once and used for multiple types or forms of visualizations, including new types of visualizations that may be developed for new technology. The electronic processor 300 may store the created visualization layers to the visualization layer database 127. Similar to the libraries and